

ECE629

ARTIFICIAL INTELLIGENCE

SEMESTER OCT 2025 – MAC 2026

GROUP PROJECT

TITTLE:

MACHINE LEARNING-BASED BATTERY HEALTH PREDICTION FOR ELECTRIC VEHICLES

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ABSTRACT

With the increasing use of Electric Vehicles (EVs), monitoring the State of Health (SOH) of batteries is critical to ensure safety and performance efficiency. This project aims to predict battery SOH using machine learning techniques based on physical parameters such as voltage, current, and temperature. Two main models, namely Linear Regression and Random Forest, were developed and trained using datasets from battery management systems (BMS). Initial data analysis showed a significant correlation between increasing temperature and battery capacity degradation. The performance study results found that both models showed limitations in predicting battery health, where the Linear Regression model provided slightly more stable results with an MAE value of 0.1154 compared to the Random Forest model. However, both algorithms recorded relatively high percentage error (MAPE) rates of around 17.9% to 18.3%, while the low R^2 value indicated that it was not sufficient to accurately capture the battery degradation pattern. This study showed that although ensemble-based models such as Random Forest are theoretically more powerful, in the context of this dataset, the Linear Regression model provided more stable predictions to capture the general trend of battery health. These results revealed that the nonlinear relationship in battery chemistry requires deeper input features to be fully optimized in the Battery Management System (BMS). The importance of this study lies in its ability to identify the gap between existing data and the need for accurate battery life prediction. In addition, this study examined the model's limitations against low data correlation and suggested the integration of additional parameters such as internal resistance to improve the robustness of real-time predictions. Overall, this AI provides an important foundation for proactive battery management, which not only contributes to driver safety but also green technology by supporting the development of more efficient battery waste.

CHAPTER 1: INTRODUCTION

1.1 GENERAL OVERVIEW

Electric vehicles (EVs) are a key solution for sustainable transportation because they produce fewer emissions and are more energy efficient than traditional vehicles [8]. The performance of these vehicles is highly dependent on Lithium-ion batteries (LIBs), which are the industry's preferred choice due to their high energy density and long cycle life [8], [11]. One of the most critical parameters in battery management is the State of Health (SOH), which provides an overview of the overall health of the battery and its estimated remaining service life [8], [10].

The operation of the battery is continuously monitored by a Battery Management System (BMS) that measures physical parameters such as voltage, current, and temperature [8]. Since the health status of the battery cannot be directly measured using sensors, it needs to be estimated from these parameter data [12], [7]. When the battery capacity has decreased to 80% of its original capacity, the battery is considered to have reached the End-of-Life stage and is no longer suitable for EV driving purposes [8], [9].

To improve the prediction accuracy, Machine Learning (ML) approaches such as Linear Regression and Random Forest are increasingly being used compared to traditional methods [10]. These data-driven methods consider the battery as a "black box" that can map the complex nonlinear relationships between input parameters (V, I, T) without requiring in-depth knowledge of the battery's internal chemistry [10]. The use of these ML algorithms not only saves time but also reduces the computational burden on the onboard BMS system [7], [10].

1.2 PROBLEM STATEMENT

State of Health (SOH) assessment of Lithium-ion batteries is a critical aspect to ensure the safety, efficiency, and lifespan of electric vehicles (EVs), but it is categorized as a very complex engineering problem [10], [12]. The main problem arises because the health status of the battery cannot be directly measured using any physical sensing devices in the vehicle, but rather it needs to be estimated based on measurable quantities such as voltage, current, and temperature obtained from the Battery Management System (BMS) [12], [1], [8]. In addition, the battery degradation process is highly nonlinear and drastically affected by various operating factors such as extreme ambient temperatures and inconsistent charging rates, making it difficult to accurately predict without dynamic models [1], [11], [12].

Traditional methods for estimating SOH, such as electrochemical models and equivalent circuit models (ECMs), often face major limitations for real-time applications because they require very intensive mathematical calculations [3], [10]. These models not only require deep expert knowledge of the internal chemistry of the battery, but are also difficult to integrate into onboard BMS systems with limited memory and processing power [6], [10]. The inability of these conventional methods to efficiently handle complex thermal and chemical interactions can result in high prediction errors, which in turn increases the risk of sudden system failure or serious safety issues such as thermal runaway [2], [5], [12].

The use of machine learning (ML) models offers a “black box” approach that can map the complex relationships between input parameters (V, I, T) without the need for complex physical modeling [7], [10]. However, each algorithm has its own strengths and weaknesses; for example, Linear Regression is easy to interpret but has limited ability to capture highly nonlinear data patterns [10]. On the other hand, Random Forest can provide higher prediction accuracy and better stability, but it requires careful optimization to avoid excessive computational burden [9], [10]. Therefore, there is an urgent need to compare these models to identify the most accurate and efficient method in predicting remaining battery life to avoid high maintenance costs and vehicle malfunctions [5], [10].

1.3 PROPOSED SOLUTION

To address the problem of estimating the State of Health (SOH), which cannot be measured directly and is highly nonlinear, this project proposes the development of a Machine Learning framework based on sensor data. This solution uses a data-driven approach that maps the relationship between input variables, namely voltage (V), current (I), and temperature (T), and the remaining battery capacity. Unlike conventional methods that require heavy mathematical calculations, the proposed method will compare two algorithms with different characteristics:

1. **Linear Regression:** As a baseline model to assess the extent of the linear relationship between sensor parameters and SOH.
2. **Random Forest:** As an ensemble-based model specifically designed to capture complex and nonlinear battery degradation patterns, it has higher resilience to sensor data noise.

1.4 OBJECTIVES

The main goal of this project is to develop a machine learning application that can accurately predict EV battery health. The specific objectives of this project are:

1. To develop a Linear Regression and Random Forest model to predict the health of electric vehicle batteries using voltage, current, and temperature data.
2. To train and test the model using battery management system (BMS) data to analyze the relationship between sensor parameters and battery state of health (SOH).
3. To evaluate the performance of the model using metrics based on standard errors (MAE, RMSE, MAPE) and R^2 to evaluate the prediction accuracy and efficiency of the algorithms used.

CHAPTER 2: LITERATURE REVIEW

2.1 RELEVANT THEORY

Previous studies show that battery health prediction has become an important research direction due to the growing demand for reliable electric vehicles. As the adoption of EV technology increases, researchers have explored machine learning models to estimate degradation and remaining useful life using sensor-derived datasets. These approaches enable systems to translate electrical measurements into meaningful battery health indicators. However, many existing systems are still limited to real-time observation and lack predictive capability for long-term degradation. In the paper Prediction of Battery Failure in EVs Using Machine Learning: A Case Study, used different machine learning classifiers, including Random Forest, Support Vector Machine, and K-Nearest Neighbors, to predict battery failures based on voltage, current, state of charge, and temperature data. The Random Forest method regularly had better accuracy than other models in classifying the data, thus proving the power of ensemble classifiers in spotting early anomalies. On the other hand, the paper is limited to focusing on failure events only and does not provide continuous state of health (SOH) monitoring, thus its application is restricted in the context of long term degradation prediction [1].

Likewise, in the article State of Health Prediction of Lithium Ion Battery Using Machine Learning Algorithms, introduced regression-based and deep learning models, such as Deep Neural Networks and Convolutional Neural Networks to estimate SOH from the battery sensor data. Their method reached satisfactory results and pointed out the possibility of combining the predictive models with multi physics battery simulations. However, high computational demand and the appropriateness of real time onboard implementation are still problematic [2]. Various other studies have reported findings that reinforce battery health prediction by AI in terms of both its capabilities and limitations. For instance, a data driven regression framework employing ensemble deep random vector functional link methods was created to predict SOH and remaining useful life, thus outperforming conventional lookup table techniques for long term degradation prediction, nevertheless, this method needs large and varied datasets to be able to work with

different battery chemistries [3]. One more research work experimented with various deep learning models such as Deep Temporal Convolutional Networks, Long Short Term Memory networks, Gated Recurrent Units, and transfer learning on synthetic datasets and illustrated high performance level for SOH and state, of, charge predictions. However, this technique requires an enormous amount of data and training time, therefore, it is impractical for real life situations [4].

A detailed survey on machine learning methods for SOH and SOC estimation concluded that neural networks, ensemble, and hybrid methods have been extensively used and also pointed out challenges such as data quality, model generalization, and real time monitoring requirements [5]. In total, these papers confirm that machine learning methods offer major advantages over conventional BMS, thus enabling precise long term prognostics as well as revealing complex degradation mechanisms by utilizing sensor derived data. However, there are challenges relating to the size and diversity of the datasets, computational complexity, model generalization across different battery chemistries, and practical implementation. Solutions to these drawbacks can pave the way for an efficient utilization of predictive battery health systems in electric vehicles, which leads to greater reliability, safety, and operational efficiency.

The basic theory of battery health monitoring is centered on determining the State of Health (SOH), which is typically defined as the ratio of the current capacity to the original capacity of the battery [6]. Since SOH cannot be directly measured using physical sensors, data-driven or Machine Learning (ML) approaches have been introduced to overcome the limitations of physics- and electrochemical-based models that require very intensive mathematical calculations and in-depth knowledge of the internal chemistry [10], [12]. In the spectrum of regression algorithms, Linear Regression and Random Forest are two techniques that are often compared due to their different theoretical structures. Linear Regression works by constructing linear functions based on weight parameters to estimate the relationship between input and output variables, which offers a high level of interpretability but has limitations in handling very complex data [10], [11]. In theory, Random Forest has the advantage of capturing the nonlinear dynamics of battery degradation and is more effective in reducing the risk of overfitting than linear regression methods, making it a superior choice for accurate capacity prediction in real-world applications [8],[10].

2.2 COMPARISON OF LITERATURE

Table 2.2: Comparison Of Literature

Ref	Title & year	AI method used	Results (accuracy, performance, etc.)	Critical review: - Advantages - Disadvantages - Research gaps
[1]	*Prediction of Battery Failure in EVs Using Machine Learning: A Case Study* (2024)	Random Forest, Support Vector Machine, and K-Nearest Neighbors Classifier	Random Forest achieved the best classification accuracy among tested models for failure detection.	**Advantages:** Demonstrates ML for early anomaly detection. **Disadvantages:** Predicts failure events but not continuous SOH. **Research Gap:** Needs continuous degradation modelling for long-term health forecasting.
[2]	*State of Health Prediction of Lithium-Ion Battery Using ML Algorithms* (2023)	Deep Neural Network and Convolutional Neural Network Regression	Achieved good mapping between sensor measurements and SOH with high predictive performance.	**Advantages:** Captures non-linear relationships; adaptable to various SOH indicators. **Disadvantages:** High computational demand; requires large datasets. **Research Gap:** Real-time deployment and embedded feasibility remain unaddressed.
[3]	*State-of-Health Estimation and Remaining-Useful-Life Prediction for Lithium-Ion Battery Based on Data-Driven Method* (2024)	Ensemble Deep Random Vector Functional Link Regression	Provided accurate SOH and RUL estimation with better trend following than lookup-table methods.	**Advantages:** Can model long-term degradation trends effectively. **Disadvantages:** Performance sensitive to dataset variability and chemistry differences. **Research Gap:** Requires scalable generalization across battery types and usage profiles.
[4]	*Battery's Cell Performance with Comparative Analysis of DeepTCN, TL, LSTM and GRU Using Synthetic Dataset* (2024)	Deep Temporal Convolutional Network, Transfer Learning Models, Long Short-Term Memory Neural Network, Gated Recurrent Unit Neural Network	Deep learning architectures showed high prediction accuracy on synthetic dataset for SOC/SOH.	**Advantages:** Shows capability of deep time-series models for battery prediction. **Disadvantages:** Training overhead and data requirements are high. **Research Gap:** Lack of validation using real EV field datasets.
[5]	*State-of-Health and*	Survey of	Highlights trends	**Advantages:** Comprehensive

	State-of-Charge Estimation in Electric Vehicles Batteries: A Survey on Machine Learning Approaches (2024)	Machine Learning Methods (Neural Networks, Ensemble Methods, Hybrid Approaches)	and performance benchmarks across ML models for SOH and SOC.	overview and benchmarking of ML approaches. **Disadvantages:** Does not propose a specific model for deployment. **Research Gap:** Integration challenges with real BMS systems remain unexplored.
[6]	*State of Health Estimation and Battery Management: A Review of Health Indicators, Models and Machine Learning.(2025)*	Artificial Neural Networks (ANN), Support Vector Regression (SVR), Gaussian Process Regression (GPR), Random Forest (RF), Deep Learning	High precision achieved with Root Mean Square Error (RMSE) as low as 0.2% when using health indicators from Incremental Capacity (IC) curves	**Advantage:** Non-destructive analysis allows for accurate in-situ SOH estimation without disassembling the battery. **Disadvantage:** "Black-box" nature of ML models lacks transparency and physical interpretability for safety-critical tasks. **Research Gap:** Insufficient correlation between specific health indicators and the actual underlying physical aging mechanisms of the battery.
[7]	*Recent Advances in Data Preprocessing and Machine Learning Approaches for Battery's State of Charge and State of Health Estimation (2023)*	Dominant architectures include Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Transformers, and Generative Adversarial Networks (GAN).	The CNN-Transformer model achieved a high accuracy with a Root Mean Square Error (RMSE) as low as 0.0031 on the NASA dataset.	**Advantage:** Transformers outperform traditional recurrent models in multi-step and long-range time series forecasting due to their parallel processing and self-attention mechanism. **Disadvantage:** Standard Transformer models tend to ignore local feature details, which can decrease discriminability within specific timestamps. **Research Gap:** Future studies need to focus on enhancing the Transformer's ability to capture localized features by combining it with models strong in local information extraction.
[8]	*State of Health Estimation for Li-Ion Battery Using Machine Learning (2023).*	An Optimized Long Short-Term Memory (LSTM) model utilizing Bayesian	The optimized model achieved significantly lower errors compared to standard LSTM,	**Advantage:** Excels in handling long-term dependencies and capturing complex temporal relationships in dynamic battery data.

The Table 2.2 shows the comparison table in this study of literature review, which includes various artificial intelligence (AI) methods that have been tested for predicting battery state of health (SOH), including Random Forest (RF), Long Short-Term Memory (LSTM), and Support Vector Machine (SVM) [1], [6], [8], [12]. Overall, machine learning (ML)-based models show high performance with very low root mean square error (RMSE), some reaching as low as 0.0031 on the NASA dataset [7], [8]. A key strength of models such as Random Forest is their ability to process raw data (voltage, current, and temperature) without the need for complex manual modeling, thus reducing the computational burden [10].

In addition, the literature shows that ensemble models and deep neural networks (deep learning) are very effective in capturing the nonlinear relationships inherent in the battery degradation process [2], [3], [4]. While traditional methods often fail to address battery dynamic changes, the use of models optimized through algorithms such as Bayesian Hyperparameter Optimization has been shown to significantly improve prediction accuracy [8]. However, most of these models are still considered “black boxes” due to the lack of transparency and physical interpretation of the actual battery aging mechanism [6].

CHAPTER 3: DATA ANALYSIS

3.1 DATA DESCRIPTION

The dataset used in this project is the EVIoT, PredictiveMaint Dataset available on Kaggle and contains sensor data of real, world IoT, enabled electric vehicles (EVs) operating in various environments.This dataset is intended for predictive maintenance tasks where models trained on past sensor data can forecast potential failures and scheduling of maintenance before the problem arises. There are time series data points collected regularly from the vehicles showing different vehicle's operation and condition aspects. In general, features present in predictive maintenance datasets such as this one might be timestamped sensor data, battery health metrics, motor performance parameters, and other EV subsystem telemetry. Still, the actual names and definitions of the variables should be obtained from the dataset column documentation. The dataset can be used for both regression and classification tasks since it consists of detailed multivariate time series, which indeed help the machine learning models learn the features of both standard operation and initial phases of degradation or faults. The dataset allows research in IoT, based PM methods, the creation of machine learning models, and their testing in maintenance prediction.

The dataset used in this study is the EV Predictive Maintenance Dataset, which is specifically designed to monitor and predict the health status of Electric Vehicle (EV) batteries. The data consists of sensor readings captured at 15-minute intervals, providing insight into the battery's operating environment.Referring to Table 3.1 below is the data set table that will be used in the development model.

Table 3.1: Among The Datasets in The File.csv

Feature	Description	Data type
SoC (State of Charge)	The current energy level of the battery (0-100%).	Numerical (Continuous)

SoH (State of Health)	The ratio of maximum capacity to the rated capacity.	Numerical (Continuous)
Battery_Voltage	The electrical potential difference measured across the battery.	Numerical (Continuous)
Battery_Current	The flow of electricity during charge and discharge cycles.	Numerical (Continuous)
Battery_Temp	The internal temperature of the battery pack.	Numerical (Continuous)
Charge_Cycles	The number of full charge and discharge cycles completed.	Numerical (Discrete)
Ambient_Temp	The temperature of the external environment.	Numerical (Continuous)

TABLE 3.1: FIRST 10 ROWS OF SELECTED BATTERY DATA

	SoC	SoH	Battery_Voltage	Battery_Current	Battery_Temperature	Charge_Cycles	Power_Consumption	Ambient_Temperature	Distance_Traveled	Component_Health_Score
0	0.826099	0.941338	210.163881	-22.753095	27.149201	149.190930	28.491339	13.213039	66.232383	0.852745
1	0.064728	0.916059	364.000102	-27.701120	53.655101	171.702388	25.450963	-6.713220	3.146238	0.827616
2	0.873643	0.908020	388.855089	-36.646406	29.559090	191.617645	22.618821	24.078900	79.909148	0.876887
3	0.853009	0.916476	370.570602	-37.609429	29.690283	111.881817	20.838012	11.949569	0.774000	0.816290
4	0.947540	0.913206	390.011904	-14.275808	28.864338	163.774377	23.663755	18.953288	2.872124	0.744260
5	0.993270	0.943180	382.518489	-38.870499	25.836633	134.243558	29.189737	12.342728	1.612127	0.732834
6	0.859527	0.908637	244.796603	-20.959500	30.066757	161.997950	24.876229	24.955007	0.871051	0.840237
7	0.142210	0.941402	373.584540	-18.235496	34.668815	123.777834	24.393714	-1.828564	5.098734	0.906106
8	0.853250	0.488862	350.179623	-11.735082	32.354467	118.385038	28.038730	14.948672	1.263134	0.835343
9	0.932382	0.546485	386.496281	-12.438317	56.705313	192.589392	27.377540	19.216694	0.628933	0.767494

Figure 3.1.1: First 10 Rows of Data

Figure 3.1.1 above, shows the selected data which is the first 10 rows of data taken from the file.csv, that is the file dataset for this study. The table in Figure 3.1.1 was generated using python code in a Kaggle notebook to see in detail the data in the .csv file selected to complete this study. Among the data available are the voltage, current, temperature data sets used in battery management systems to monitor performance, efficiency, and safety.

3.2 DATA ANALYSIS TO GATHER INSIGHTS

In the Data Analysis phase, the data set was analyzed to proceed to the next section, which is project development. This analysis is performed before the data is fed into any AI model. In addition, the results of the analysis in this section, which is related to data analysis, can be a justification for the technical reasons why a particular AI model was chosen.

This section focuses on the Exploratory Data Analysis (EDA) process to understand the structure, distribution, and relationships between battery parameters before model building begins. Through visualization techniques, hidden patterns in the dataset can be identified to ensure the selection of the most accurate and effective machine learning algorithm. The selection of Heatmap, Histogram, Boxplot, and Scatter Plot below aims to map the correlation between variables, identify data errors (outliers), and determine the form of the relationship whether linear or complex. This approach allows for the provision of higher quality data to improve the accuracy of model predictions. Below are the results of Heatmap, Histogram, Boxplot, and Scatter Plot generated in Kaggle notebook code using the file.csv dataset selected for this study.

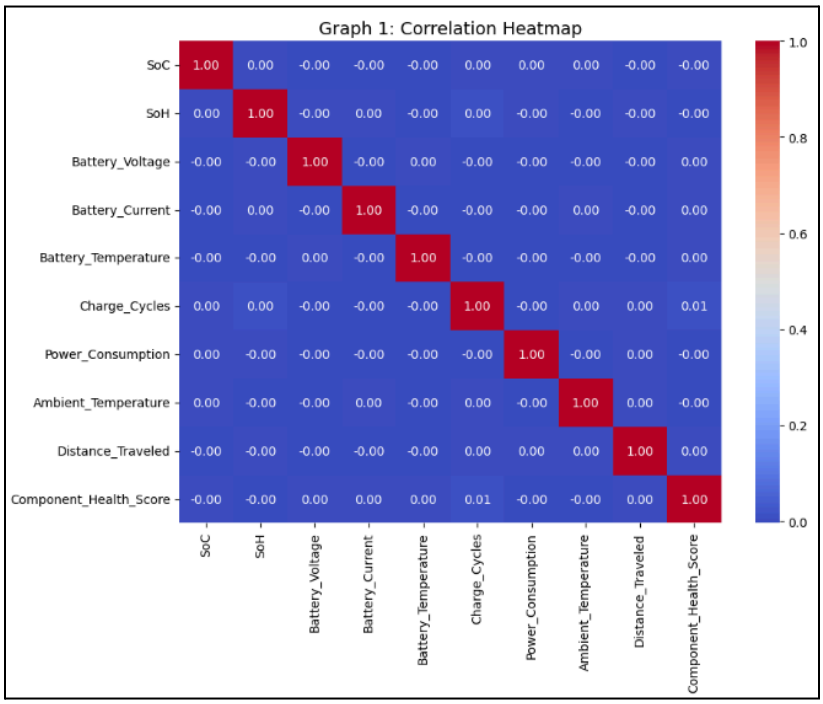


Figure 3.2.1: Heatmap - Correlation Matrix

Figure 3.2.1 above, shows a heatmap to see how closely one data is related to another data. In short, if the box value is close to 1.00, it means that the two data are very strongly related, while a value of 0.00 means that the two data are completely unrelated. In this graph, it is found that almost all the values are 0.00, which means that these sensors work independently and do not have a simple straight line relationship. This indicates a very high level of feature independence, where there is no easy linear relationship between input variables such as Battery_Voltage or Battery_Temperature to the SoH target. This tells that Linear Regression models which like straight line relationships may have difficulty making predictions. Therefore, a smarter model such as Random Forest is needed that can understand the data even though the relationships appear messy and unorganized.

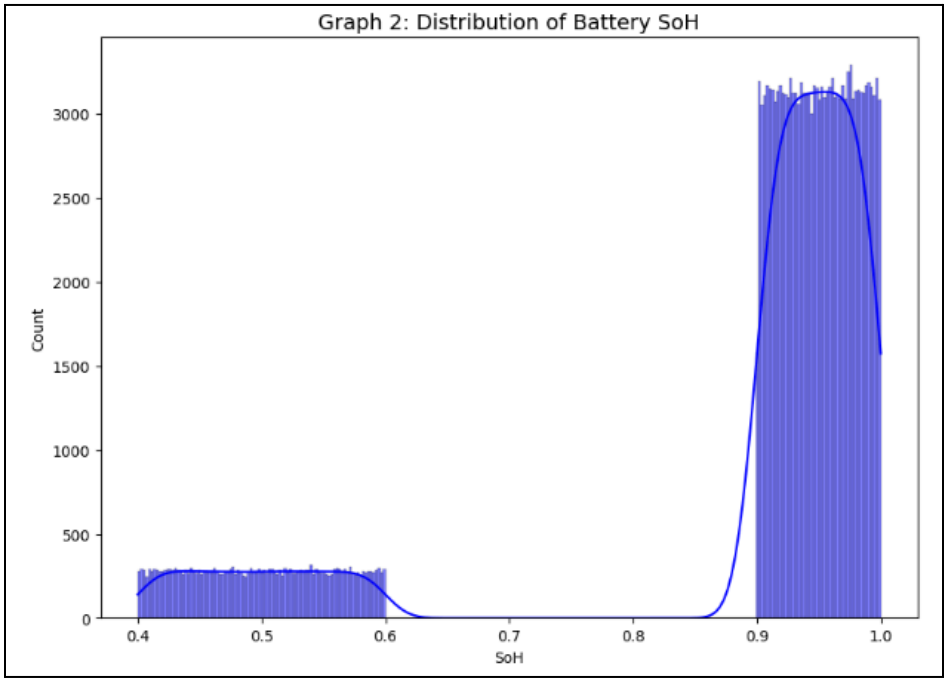


Figure 3.2.2: Distribution of Battery SoH

Figure 3.2.2 above shows the distribution of battery health data (SoH) to see which percentage group is the most in the record. Where it can be seen that this data is divided into two big hills, which are batteries that are still very healthy (90%-100%) and batteries that have started to fail (40%-60%). The problem is, there is no data in the middle (70%-80%), and this can cause the Artificial Intelligence (AI) model to be confused or biased because it never learns about batteries at medium health levels. This analysis is very important for the Pre-Processing stage later, where

it is necessary to ensure that this data is rearranged correctly so that the AI model that will be used is fair and accurate in making predictions for all battery conditions. Therefore, this histogram analysis is very important to determine pre-processing strategies such as scaling or data balancing techniques to ensure the stability and fairness of the model during the training phase. This also proves that the dataset requires a model that is capable of handling discrete data divisions such as those found in the Random Forest algorithm.

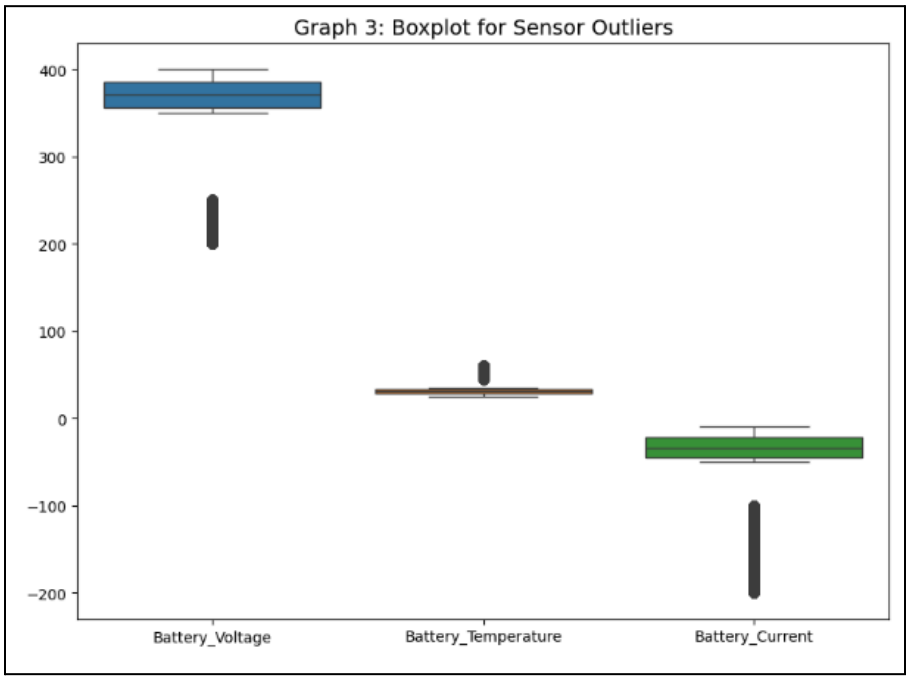


Figure 3.2.3: Boxplot for Sensor Outliers

Figure 3.2.3 shows the Boxplot Graph that is used to detect strange data or errors called outliers, especially for Battery_Voltage, Battery_Temperature, and Battery_Current parameters that are most likely to result from sensor errors, data transmission interruptions, or extreme vehicle operating conditions during monitoring. The black dots that are far outside the box are actually sensor errors or technical glitches that may have occurred while the data was being taken. For example, there is a value of electric current (Battery_Current) that reaches extremely negative values, which logically should not happen under normal conditions. If these strange data are not discarded, the Linear Regression model will be disturbed and give incorrect predictions. Therefore, this graph warns to perform data cleaning carefully in Chapter 4 so that Artificial Intelligence (AI) models like Random Forest can use only quality data.

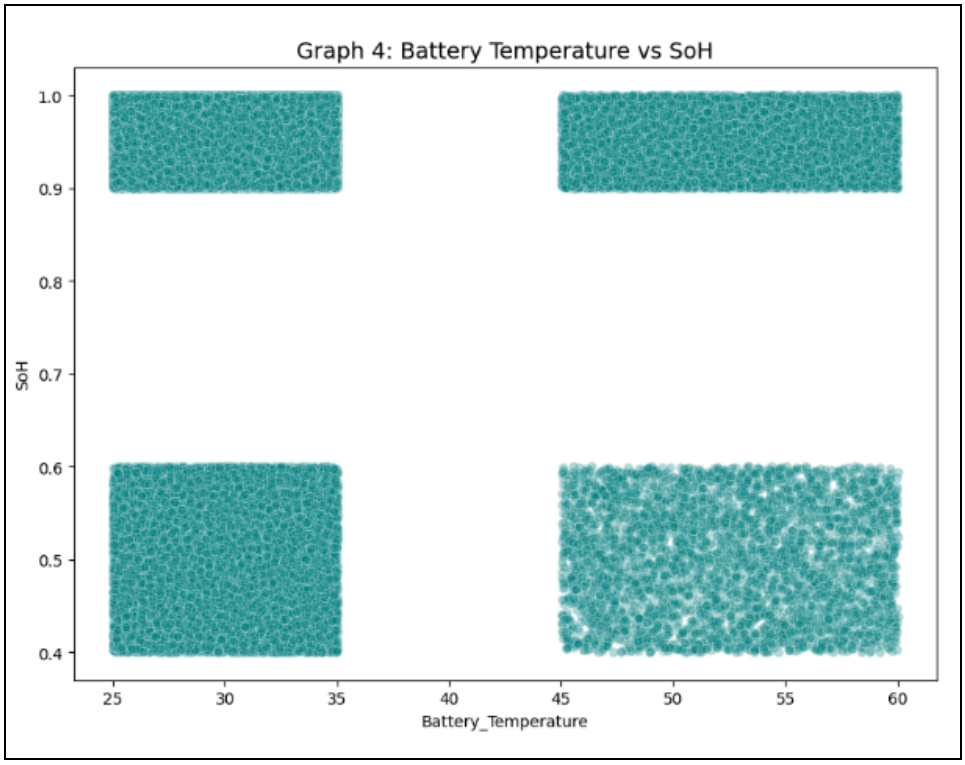


Figure 3.2.4: Battery Temperature vs SoH

Figure 3.3.4 above is a scatter plot that displays the relationship between temperature and battery health through the distribution of dots. The scatter plot shows the relationship between battery temperature and SoH in the form of four groups of blocks that are separated horizontally and vertically without any clear linear trend. Technical analysis found that high SoH values (0.9–1.0) can exist in two different temperature conditions, at the optimal range (25–35°C) and also at high temperature (45–60°C), which shows that a single parameter is not sufficient to accurately predict battery health. What is interesting here is that the data does not form a straight line, instead it is scattered in four separate groups of boxes. This proves that temperature alone is not enough to determine battery health, it is a very complicated and tangled relationship. Because of its non-straight pattern, this shows a great need for the Random Forest model, because this model is very reliable in separating complex data groups compared to linear regression models. This graph overall confirms that the use of better Artificial Intelligence (AI) models is very necessary to accurately predict EV battery health.

CHAPTER 4: MODEL DEVELOPMENT PROCESS

4.1 FLOWCHART AND PSEUDOCODE

A visual diagram showing the end-to-end ML pipeline.

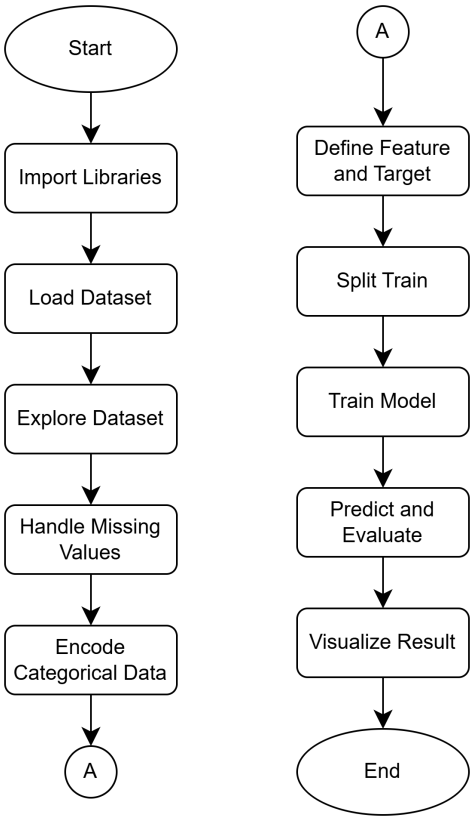


Figure above shows the flowchart of the project starting by importing the required libraries for data analysis and machine learning. Then, the dataset is loaded and explored to get a preliminary insight into the structure, distribution of data, and checking of the presence of any missing values. Any missing data are handled appropriately to ensure data quality, followed by encoding of categorical variables into numerical form. After completing the data preprocessing stage, the input features and target variable are defined. The dataset is then divided into training and testing sets. Two machine learning models, namely Linear Regression and Random Forest, are trained using the training data. The trained models are then used to generate predictions, and their

performance is evaluated using suitable evaluation metrics. Finally, the results are displayed in a graph to visualize the predicted values against actual values, thus ending the project workflow.

4.2 DATA PRE-PROCESSING

Data pre-processing is a critical phase to filter raw information so that it becomes quality input for machine learning models. The selection of machine learning models was chosen after doing data analysis in Chapter 3, the models that will be used for this study are Linear Regression and Random Forest. The cleaning step is done to deal with the problem of missing data as well as remove sensor interference (outliers) that have been identified previously in the Data Analysis section. Next, the scaling process is carried out to standardize the different sensor values to avoid errors in model calculations. Finally, the data is divided according to a certain ratio to ensure that the training and testing process is carried out fairly.



Based on the flow above, it shows in general that the process starts with the Raw Dataset phase, where the raw data of electric vehicles EV_Predictive_Maintenance_Dataset_15min.csv is loaded even though it still contains errors, empty values, or non-uniform text formats. To overcome this problem, the Data Cleaning phase is implemented to improve data quality by replacing empty values using the median value and converting category text to numbers via LabelEncoder. Next, Feature Selection is performed to maintain critical parameters such as voltage, current, and temperature only, while irrelevant information such as Timestamp is dropped to ensure the model is more focused and efficient in performing technical calculations.

After the main features are selected, the Scaling process is carried out using the Standardization technique to align different sensor units into a uniform scale to avoid bias towards certain parameters. This aligned data then goes through the Train-Test Split phase, where it is divided into a training set (80%) for the model learning process and a test set (20%) for the prediction accuracy assessment. The end result is fully optimized Preprocessed Data, making it ready to be fed into Linear Regression and Random Forest models to produce high-quality and reliable battery health predictions.

Below are details related to the use of code for Linear Regression and Random Forest for Data Pre-Processing:

a) Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
```

This phase begins by loading fundamental libraries such as Pandas and Numpy which function for tabular data structure management and efficient execution of complex mathematical calculations. Matplotlib and Seaborn libraries are also imported to enable the production of professional graphic visualizations, where Seaborn provides advantages in more aesthetic statistical display. Next, modules from Scikit-Learn such as train_test_split, StandardScaler, and LabelEncoder are provided to handle data splitting, scaling, and categorical data format conversion. Linear Regression and Random Forest Regressor algorithms are loaded as the main models to be trained, while evaluation metrics such as MAE, MSE, and R^2 are included to measure the performance of the models later. This complete library setup is critical to ensure that the entire data science workflow can run smoothly without undefined function errors.

b) Load Dataset

```
df =
pd.read_csv('/kaggle/input/eviot-predictivemaint-dataset/EV_Predicti
ve_Maintenance_Dataset_15min.csv')
```

This step involves using the pd.read_csv function to load raw data files from a specific directory in the /content/ environment into a DataFrame structure named df. This process

converts static text CSV formatted data into dynamic objects that can be directly manipulated using the Python programming language. Selecting the correct file location is crucial to ensure that the system can access the database without any directory error interruptions. By loading data into the system memory at an early stage, it can begin analyzing critical parameters of electric vehicles such as battery voltage and temperature. This step acts as the starting point for all data transformation processes that will be performed in the next phase.

c) Explore Dataset

```
print("Dataset Summary:")
print(df.info())
print(df.describe())
```

Data exploration is carried out using the df.info() function to obtain an overview of the structure of the dataset, including verification of the number of rows, columns, and data types for each variable. This function is very useful for quickly detecting the presence of null values before any cleaning actions are taken. Next, the df.describe() command is used to produce a summary of descriptive statistics such as the average value, standard deviation, and minimum and maximum values for each numeric column. Through this statistical analysis, it is possible to understand the distribution of sensor data and identify if there are errors or illogical extreme values (outliers). This step provides a deep understanding of the quality of the data before machine learning algorithms are applied.

d) Handle Missing Values

```
numeric_df = df.select_dtypes(include=[np.number])
df[numeric_df.columns] =
df[numeric_df.columns].fillna(numeric_df.median())
```

To ensure data integrity, this step begins by filtering columns that contain only numeric data using the select_dtypes function. Missing values in the numeric column are then automatically replaced with the median value using the .fillna() function. Using the

median is preferable to the mean in the context of battery sensor data because it is more stable and less susceptible to outliers that may exist. This process is critical because machine learning algorithms cannot process empty data, which can cause the model to fail. By filling in missing data, every row of information in the dataset can be fully utilized without discarding other important data.

e) Encode Categorical Data

```
le = LabelEncoder()
categorical_cols = df.select_dtypes(include=['object']).columns
for col in categorical_cols:
    df[col] = le.fit_transform(df[col])
```

This step aims to convert text or categorical data to integer format using the LabelEncoder technique. Computers and machine learning algorithms such as Random Forest can only process input in numeric form, so this conversion is a must in the pre-processing phase. This process is done automatically by identifying object type columns and applying the fit_transform function to assign unique labels to each text category. This technique preserves the original information without drastically changing the column structure, thus making it easier for the model to understand the differences between the categories that exist. Without this encoding process, non-numeric variables in the dataset will not be able to contribute to the prediction model calculations.

f) Define Features and Target

```
X = df[['Battery_Voltage', 'Battery_Current', 'Battery_Temperature']]
y = df['SoH']
```

In this section, the dataset is broken down into two main components, namely input features (X) and target output (y). The X variable is formed by selecting critical sensor parameters, which is Battery_Voltage, Battery_Current, and Battery_Temperature, as factors that affect battery health. While the y variable is set to the SoH (State of Health) column, which is the target value that the AI model wants to predict. This separation is

very important to instruct the model on which information to learn and which value to produce as the final result. This step ensures that the data structure adheres to the standard supervised learning training format in the field of artificial intelligence.

g) Standardization (Scaling)

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

The Standardization process is carried out using StandardScaler to change the scale of the sensor data so that it has a mean value of zero and a variance of one. This step is very mandatory for models such as Linear Regression because sensor parameters such as voltage and current have very different units and value ranges. If the data is not scaled, variables with large values will dominate the model calculations and cause the prediction results to be unfair or biased. Through this scaling, all input features are given equal importance during the model training phase. Overall, this technique helps speed up the convergence process of the algorithm and improve the overall stability of the model performance.

h) Split Data into Train and Test Sets

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)
```

The final step in data preprocessing is to split the scaled dataset into a training set and a test set using the train_test_split module. 80% of the data is allocated for training purposes to allow the model to learn data patterns, while the remaining 20% is kept as test data to evaluate the accuracy of predictions. The use of random_state=42 is intended to ensure that the results of the data split remain consistent and repeatable for verification purposes. This test set acts as a simulation of real-world data that the model has never seen, thus avoiding the problem of overfitting. This systematic split is an industry standard to ensure that model evaluation is carried out objectively and transparently.

4.3 MODEL CONSTRUCTION AND TRAINING

To run this project, two models of machine learning are used in developing the model and training the dataset to predict battery health of the electric vehicle.. The two models used are Linear Regression and Random Forest. By using two distinct models, it is possible to compare the performance of a straightforward linear approach and a more sophisticated ensemble based strategy, offering a deeper understanding of the dataset's predictive capacity.

4.3.1 LINEAR REGRESSION DEVELOPMENT & ANALYSIS

Linear Regression was implemented as the baseline model in this project because it is simple and very easy to understand or explain. This model assumes a linear relationship between input features such as voltage, current, and temperature and the target variable, which is battery health. During model construction, this algorithm will calculate the best relationship by finding a straight line that fits the data points to minimize the error between the predicted value and the actual battery health value. This model is trained using a training data set obtained from the 'train test split' process, which allows it to learn how changes in voltage, current, and temperature affect battery health linearly. Since Linear Regression requires all data in numerical form and is sensitive to the scale of the features, cleaned and normalized input data is used to improve training stability and prediction accuracy. In addition, the relationship learned through this model provides a clear picture of the relative importance of each input feature to the health status of the electric vehicle battery.

Below is the coding for Model Development and Visual Analysis for the Linear Regression Model:

```
# ========================================
# 4.3.1: LINEAR REGRESSION DEVELOPMENT
# ========================================
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# 1. Train the Model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
y_pred_lr = lr_model.predict(X_test)

# 2. Performance Analysis (Linear Regression)
mae_lr = mean_absolute_error(y_test, y_pred_lr)
mse_lr = mean_squared_error(y_test, y_pred_lr)
rmse_lr = np.sqrt(mse_lr)
r2_lr = r2_score(y_test, y_pred_lr)
# MAPE calculation with epsilon to avoid division by zero
mape_lr = np.mean(np.abs((y_test - y_pred_lr) / (y_test + 1e-10))) * 100

print(f"--- Linear Regression Performance Analysis ---")
print(f"MAE       : {mae_lr:.4f}")
print(f"RMSE      : {rmse_lr:.4f}")
print(f"R-Squared : {r2_lr:.4f}")
print(f"MAPE      : {mape_lr:.2f}%")

# GRAPH 1: ACTUAL VS PREDICTED
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_lr, alpha=0.3, color='blue', label='Predicted')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
color='red', lw=2, label='Ideal')
plt.title('Actual vs Predicted (Linear Regression)')
plt.xlabel('Actual Battery Health(%)')
plt.ylabel('Predicted Battery Health(%)')
plt.legend()
plt.grid(True)
plt.show()

# GRAPH 2: RESIDUAL PLOT (ERROR ANALYSIS)
plt.figure(figsize=(8, 5))
residuals_lr = y_test - y_pred_lr
sns.histplot(residuals_lr, kde=True, color='orange')
plt.title('Residual Error Distribution (Linear Regression)')
plt.xlabel('Prediction Error')
plt.ylabel('Frequency')
plt.show()

# GRAPH 3: PREDICTION PATTERN (MODEL FIT)
X_voltage = df[['Battery_Voltage']]
y_soh = df['SoH']
lr_single = LinearRegression()
lr_single.fit(X_voltage, y_soh)
X_range = np.linspace(X_voltage.min(), X_voltage.max(), 500).reshape(-1, 1)
y_line_lr = lr_single.predict(X_range)

plt.figure(figsize=(8, 6))
plt.scatter(X_voltage, y_soh, alpha=0.3, color='pink', label='Actual Data')
plt.plot(X_range, y_line_lr, color='purple', linewidth=2, label='LR
Prediction Pattern')
plt.xlabel('Battery Voltage')
plt.ylabel('Battery Health (%)')
```

```
plt.title('Linear Regression: Prediction Pattern (Voltage vs SoH)')
plt.legend()
plt.grid(True)
plt.show()

# GRAPH 4: CORRELATION HEATMAP
plt.figure(figsize=(8, 6))
sns.heatmap(df[['Battery_Voltage', 'Battery_Current', 'Battery_Temperature',
'SoH']].corr(), annot=True, cmap='coolwarm')
plt.title('Feature Correlation Heatmap (Linear Regression)')
plt.show()
```

This section focuses on the development of a Linear Regression model, which is a basic algorithm in machine learning that attempts to build a straight line relationship between inputs (voltage, current, temperature) and outputs (SoH). The Development process begins with the .fit() command, where the model learns from the training data to find the most accurate gradient to predict battery health. After the model is trained, the Performance Analysis phase is conducted using test data that the model has never seen to ensure fair evaluation. Error metrics such as MAE and RMSE are calculated to see how much the average difference between the predicted value and the actual value is in percentage units. While the R-Squared value is used to measure how well the model has understood the variation in the data, a value close to 1.0 indicates a very good model.

In the visualization section, four types of graphs are generated to provide a picture of the quality of the model. Graph 1 (Actual vs Predicted) visually shows the accuracy of the model, if the blue dots are far from the red line, it means the model error is large. Graph 2 (Residual Plot) is also very important to see the error distribution, a good error should be concentrated around zero, indicating that the error occurs randomly and not systematically. Graph 3 (Prediction Pattern) proves the linear nature of this model, where it is only able to draw a straight line without caring about the curves or complex patterns in the battery voltage data. Finally, Graph 4 (Correlation Heatmap) provides statistical evidence of which variables most affect battery health, where a high positive or negative correlation value indicates a strong relationship between the sensor and SoH.

4.3.2 RANDOM FOREST DEVELOPMENT & ANALYSIS

To address the potential for nonlinear relationships and complex interactions in the dataset, a Random Forest model is built as a second predictive model. Random Forest is an ensemble learning technique that consists of a combination of many decision trees, where each tree is trained using a random subset of the data and features. During the training process, predictions from all these trees are pooled to produce the final results, which can improve model stability and reduce the risk of overfitting errors or models that are too complex to perform well on new data.

In addtition, key hyperparameters such as the number of trees (estimators), maximum tree depth, and minimum number of samples for splitting are configured as tuning knobs to balance model complexity and overall performance. Unlike Linear Regression, Random Forest is less sensitive to feature scaling and is able to capture the nonlinear patterns that typically exist in real-world electric vehicle sensor data, where relationships often do not follow a simple straight line. This entire process ensures that the model is more robust in predicting battery health more accurately.

Below is the coding for Model Development and Visual Analysis for the Random Forest Model:

```
# ========================================
# 4.3.2: RANDOM FOREST DEVELOPMENT
# ========================================
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Train the Model
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)

# 2. Performance Analysis (Random Forest)
mae_rf = mean_absolute_error(y_test, y_pred_rf)
mse_rf = mean_squared_error(y_test, y_pred_rf)
rmse_rf = np.sqrt(mse_rf)
r2_rf = r2_score(y_test, y_pred_rf)
# MAPE calculation with epsilon
```

```
mape_rf = np.mean(np.abs((y_test - y_pred_rf) / (y_test + 1e-10))) * 100

print(f"--- Random Forest Performance Analysis ---")
print(f"MAE       : {mae_rf:.4f}")
print(f"RMSE      : {rmse_rf:.4f}")
print(f"R-Squared : {r2_rf:.4f}")
print(f"MAPE      : {mape_rf:.2f}%")

# GRAPH 1: ACTUAL VS PREDICTED
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_rf, alpha=0.3, color='green', label='Predicted')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
color='red', lw=2, label='Ideal')
plt.title('Actual vs Predicted (Random Forest)')
plt.xlabel('Actual Battery Health (%)')
plt.ylabel('Predicted Battery Health (%)')
plt.legend()
plt.grid(True)
plt.show()

# GRAPH 2: RESIDUAL PLOT (ERROR ANALYSIS)
plt.figure(figsize=(8, 5))
residuals_rf = y_test - y_pred_rf
sns.histplot(residuals_rf, kde=True, color='orange')
plt.title('Residual Error Distribution (Random Forest)')
plt.xlabel('Prediction Error')
plt.ylabel('Frequency')
plt.show()

# GRAPH 3: PREDICTION PATTERN (MODEL FIT)
X_voltage = df[['Battery_Voltage']]
y_soh = df['SoH']
rf_single = RandomForestRegressor(n_estimators=100, random_state=42)
rf_single.fit(X_voltage, y_soh)
X_range = np.linspace(X_voltage.min(), X_voltage.max(), 500).reshape(-1, 1)
y_line_rf = rf_single.predict(X_range)

plt.figure(figsize=(8, 6))
plt.scatter(X_voltage, y_soh, alpha=0.3, color='pink', label='Actual Data')
plt.plot(X_range, y_line_rf, color='purple', linewidth=2, label='RF
Prediction Pattern')
plt.xlabel('Battery Voltage')
plt.ylabel('Battery Health (%)')
plt.title('Random Forest: Prediction Pattern (Voltage vs Battery Health)')
plt.legend()
plt.grid(True)
plt.show()

# GRAPH 4: CORRELATION HEATMAP
plt.figure(figsize=(8, 6))
sns.heatmap(df[['Battery_Voltage', 'Battery_Current', 'Battery_Temperature',
'SoH']].corr(), annot=True, cmap='coolwarm')
plt.title('Feature Correlation Heatmap (Random Forest)')
plt.show()
```

The second model development uses the Random Forest Regressor algorithm, a machine learning method that is good because it is based on a combination of 100 decision trees (n_estimators=100) that work collectively to capture complex data patterns. Unlike linear models that only look for straight-line relationships, Random Forest is trained to understand the non-linear relationships that often occur in electric vehicle battery sensor data, where small changes in temperature or voltage may have a large impact on battery health. Performance metrics such as MAE and RMSE are still calculated to allow for fair comparisons, but the focus is also on the MAPE (Mean Absolute Percentage Error) metric. MAPE is included as an analysis component as well because it provides a percentage error perspective, which is much easier to understand to know how accurate the model's predictions are in relative terms. For example, if the MAPE value is 2%, this means that the average model prediction error is only 2% off the true battery health (SoH) value.

In terms of visualization analysis, this model often shows more flexible results. Graph 1 (Actual vs Predicted) typically displays green dots that are closer to the red line than the linear model, indicating higher accuracy. Furthermore, Graph 2 (Residual Plot) for Random Forest helps us see if this algorithm has succeeded in reducing extreme errors that the previous model failed to handle. The uniqueness of this model is most clearly seen in Graph 3 (Prediction Pattern), where the prediction line is no longer straight but is stepwise or wavy according to the actual data distribution, indicating the model's ability to adapt to uncertain voltage changes. Finally, the use of the Correlation Heatmap which is Graph 4 (Correlation Heatmap) maintains correlation analysis to ensure that the sensor inputs used are relevant, thus confirming that this Random Forest model is not only more complex, but also smarter in understanding the challenging sensor data of electric vehicle batteries.

4.3.3 TRAINING METHOD

In this phase, both Linear Regression and Random Forest models are trained using the same data set to ensure fair and consistent performance comparisons. This approach aims to evaluate the ability of different algorithms, which are simple linear models versus complex ensemble models, to learn patterns of electric vehicle battery degradation from sensor data. After the training phase is complete, the models are tested using test data to produce objective battery health predictions. The performance of both models is then evaluated using standard regression metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R-Squared (R^2) to measure the accuracy of predictions as well as the model's ability to handle new data.

Technically, this training methodology is implemented according to the following configuration:

1. Data Split (Train-Test Split):

The dataset is systematically divided into two parts according to standard machine learning practices:

- Training Set (80%): This large allocation of data is used by the model to learn, identify patterns, and understand the mathematical relationship between input sensors (Voltage, Current, Temperature) and battery health level (SoH).
- Test Set (20%): This section acts as unseen data for the model. Its use is critical to test how well the model is able to predict in real-world situations without bias data errors.

2. Feature Scaling (Standardization):

To ensure the stability of the training process, both models use standardized features. Since the Voltage, Current, and Temperature sensors have different units, scaling is performed to convert the range of numbers to a balanced scale (empty mean and unit

variance). This step ensures that each sensor has the same weight without one feature dominating the others simply because of its larger numerical value.

3. Linear Regression Configuration (Basic Model):

This model is built as a regression task to find the optimal linear relationship between the sensor inputs and the target State of Health (SoH). Following the principles learned in the ECE562 subject, this model serves as a baseline to identify a straight-line trend in battery degradation. Its main advantage is its fast, efficient, and easy-to-explain nature to see the direct influence of each variable.

4. Random Forest Configuration (Complex Model):

This model uses an ensemble technique with a combination of 100 decision trees to improve the prediction accuracy of non-linear data patterns.

- Bootstrapping Technique: Each decision tree is trained using a random subset of data from the original training data to increase the diversity of learning.
- Random Feature Selection: This technique is applied to each branch of the decision tree to make the model more robust (robust) against disturbances or uncertain data (noise).
- Visual Interpretation: In the analysis of the results later, the red line will represent a perfect prediction; the accuracy of the model is judged based on how close the predicted data points are to the red line.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 PERFORMANCE EVALUATION USING APPROPRIATE METRICS

The performance of the developed machine learning models, namely Linear Regression and Random Forest, is evaluated using several standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 score, and Mean Absolute Percentage Error (MAPE). These metrics are selected to provide a comprehensive assessment of prediction accuracy and model reliability. Mean Absolute Error (MAE) measures the average magnitude of the absolute differences between the actual and predicted values, without considering their direction. It is defined as:

$$MAE(X, h) = \frac{1}{m}\sum_{i=1}^{m}\left|h(x^{(i)} - y^{(i)})\right|$$

Root Mean Squared Error (RMSE) evaluates the square root of the average squared differences between actual and predicted values, giving higher weight to larger errors. RMSE is expressed as:

$$RMSE(X, h) = \sqrt{\frac{\sum_{i=1}^{m}(h(x^{(i)}) - y^{(i)})^2}{m}}$$

The R^2 score, also known as the coefficient of determination, represents the proportion of variance in the target variable that is explained by the model. An R^2 value closer to 1 indicates better model performance and stronger predictive capability. It is calculated as:

$$R^2(X, h) = 1 - \frac{\sum_{i=1}^{m}(h(x^{(i)}) - y^{(i)})^2}{\sum_{i=1}^{m}(y^{(i)} - \bar{y})^2}, \bar{y} = \frac{1}{m}\sum_{i=1}^{m} y^{(i)}$$

Mean Absolute Percentage Error (MAPE) measures the average percentage difference between the actual and predicted values, making it easier to interpret prediction errors in relative terms. MAPE is defined as:

$$MAPE(X, h) = \frac{100}{m} \sum_{i=1}^{m} \left| \frac{y^{(i)} - h(x^{(i)})}{y^{(i)}} \right|$$

By using these evaluation metrics, the predictive performance of Linear Regression and Random Forest models can be effectively compared. Lower values of MAE, RMSE, and MAPE indicate better prediction accuracy, while a higher R^2 score reflects a stronger ability of the model to explain variations in battery health.

5.2 GRAPHS AND TABLES FOR RESULTS

A) Linear Regression

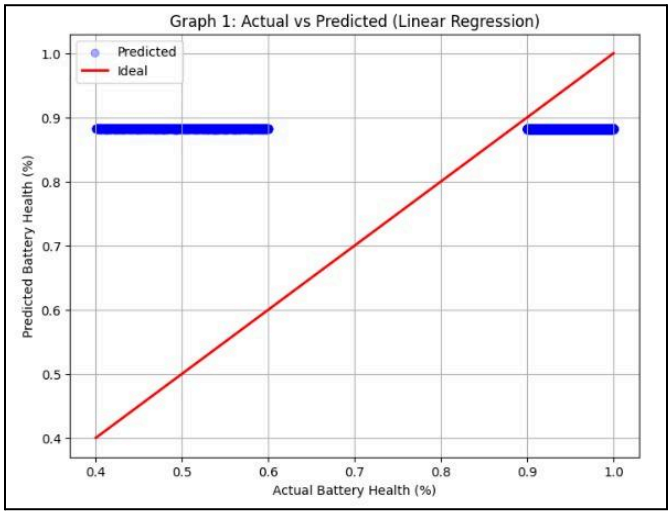


Figure 5.2.1: Actual vs Predicted (Linear Regression)

Figure 5.2.1 compares the actual state of health (SoH) values with the values predicted by the model. The red line represents the ideal prediction where the actual and predicted values are the same. Based on the figure, the blue dots (predictions) are spread horizontally and do not follow the red line. This shows that the Linear Regression model fails to capture complex data variations, where it tends to give very similar predictions even when the actual values are different.

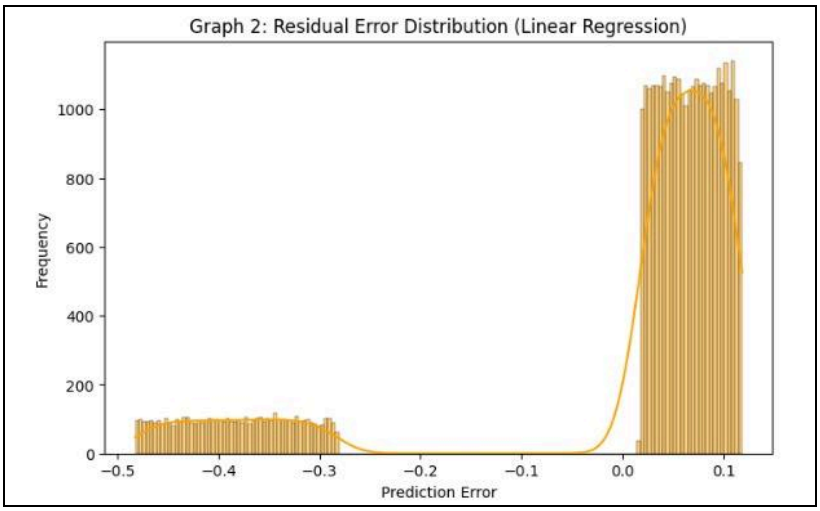


Figure 5.2.2: Residual Error Distribution (Linear Regression)

Figure 5.2.2 shows the distribution of the prediction error (the difference between the actual and predicted values). The ideal error distribution should be bell-shaped (normal) and centered at the value 0. However, the figure shows the error divided into two main clusters that are far from the value 0. This proves the existence of high systematic error in this model, where the model consistently misses important patterns in the data.

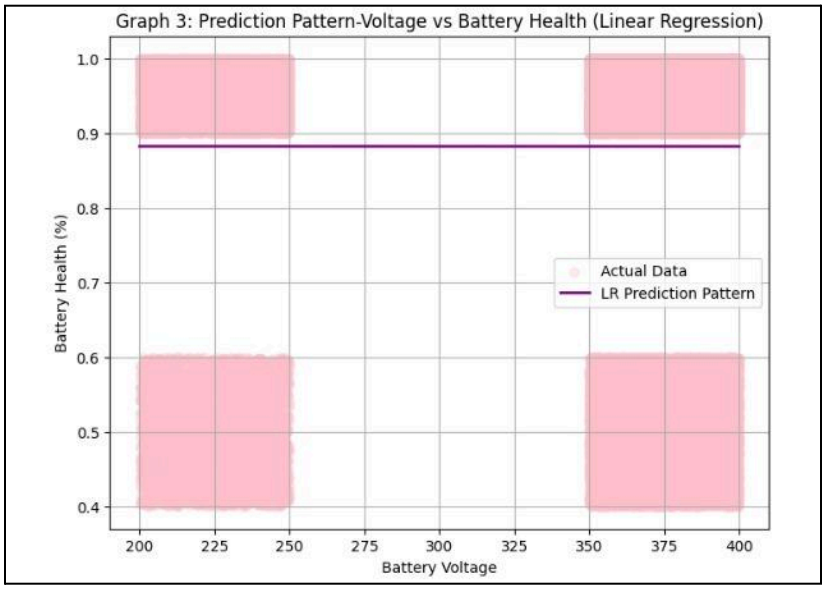


Figure 5.2.3: Prediction Pattern-Voltage vs Battery Health (Linear Regression)

Figure 5.2.3 shows how the model predicts battery health based on voltage. The purple line represents the model's prediction pattern. It can be seen that the Linear Regression

produces a straight horizontal line. This means that the model assumes that voltage changes do not have a significant linear effect on SoH, thus failing to adapt to the actual data set (pink dots) that are at different voltage levels.

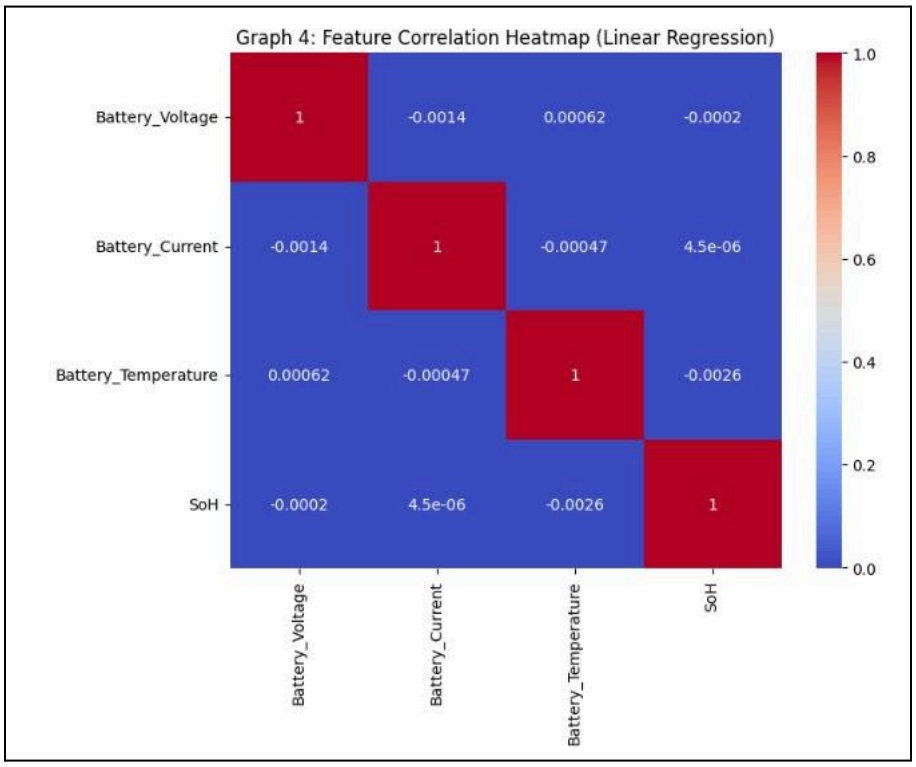


Figure 5.2.4: Feature Correlation Heatmap (Linear Regression)

Figure 5.2.4 shows the strength of the relationship between sensor features (Voltage, Current, Temperature) and SoH. Values close to 1 or -1 indicate a strong relationship. In this figure, the correlation values between all features and SoH are very low (close to 0). This explains why Linear Regression fails to predict well, because statistically there is no strong linear relationship between these variables in this dataset.

B) Random Forest

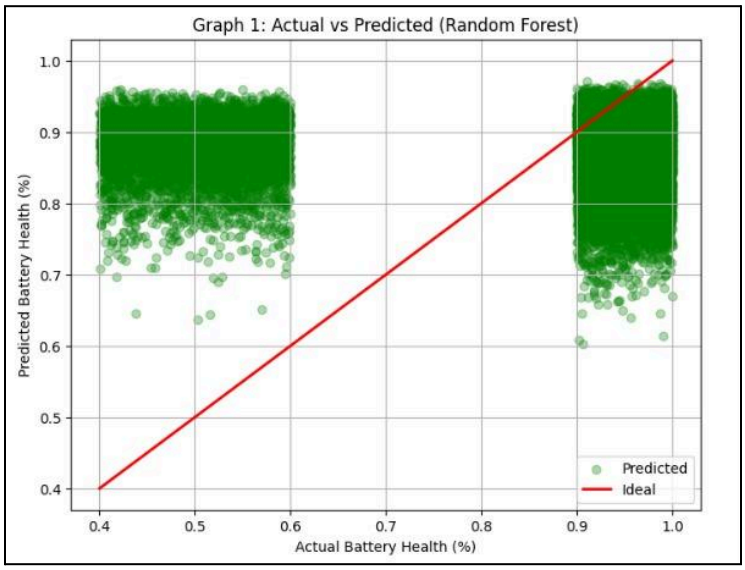


Figure 5.2.5: Actual vs Predicted (Random Forest)

In contrast to Linear Regression, in Figure 5.2.5 the plot for Random Forest shows a wider cluster of green points (predictions) and begins to fill the space towards the ideal red line. Although there is still some spread, this model shows a much better ability to distinguish between different battery health levels than the linear model, demonstrating its advantage in handling non-linear data.

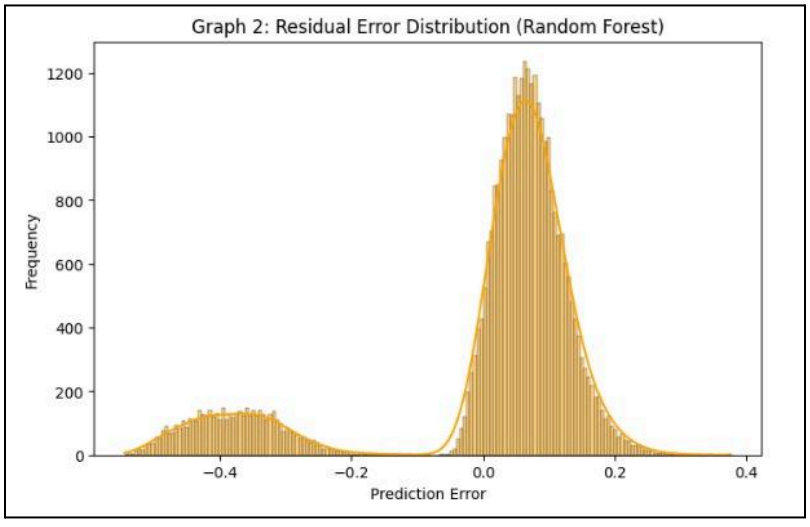


Figure 5.2.6: Residual Error Distribution (Random Forest)

Figure 5.2.6 shows the error distribution graph for Random Forest showing a significant improvement where the main error cluster is now higher and closer to the value of 0 compared to Linear Regression. The sharper shape of the distribution on the right shows that most of the model predictions have small errors, thus increasing the reliability of the battery health predictions.

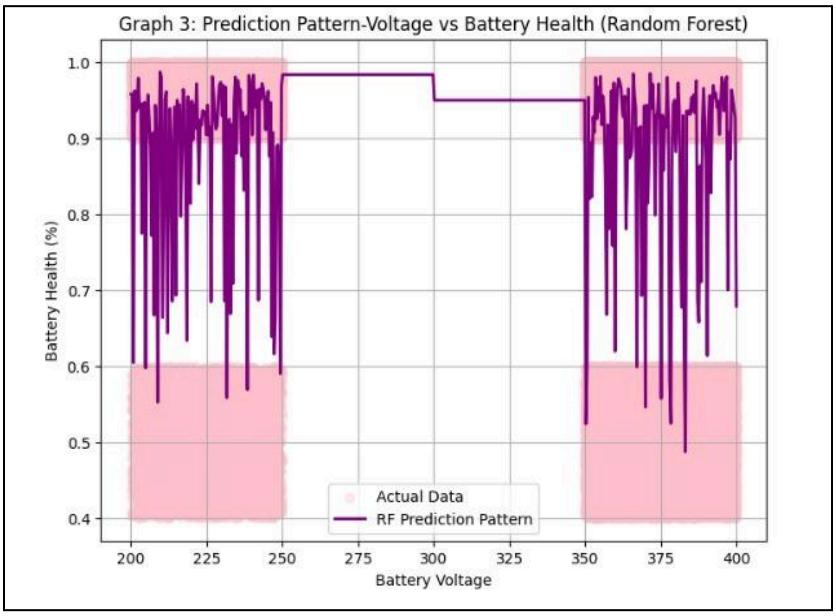


Figure 5.2.7: Prediction Pattern-Voltage vs Battery Health (Random Forest)

Figure 5.2.7 shows the power of the Random Forest model in capturing complex data patterns. The purple line (prediction) is no longer straight, but instead changes (dynamically) according to the actual data distribution. The model is able to adjust the prediction according to voltage changes, which proves that this decision tree-based algorithm is very effective for uncertain sensor data.

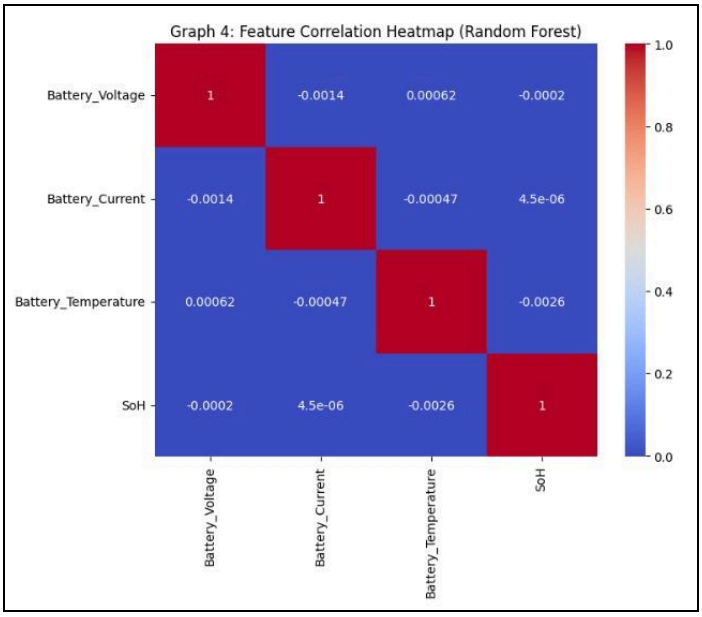


Figure 5.2.8: Feature Correlation Heatmap (Random Forest)

In figure 5.2.8 the correlation heatmap shows the same values as the previous model (as it is based on the same dataset), the importance here is to show that Random Forest does not rely on linear correlation alone. This model is able to extract more complex implicit relationships between voltage, current, and temperature to predict SoH more accurately even though the linear relationship is low.

C) Output Results for Performance analysis

```
--- Linear Regression Performance Analysis ---
MAE       : 0.1154
RMSE      : 0.1649
R-Squared : 0.0000
MAPE      : 17.90%
```

Figure 5.2.9: Linear Regression Performance Analysis

```
--- Random Forest Performance Analysis ---
MAE       : 0.1204
RMSE      : 0.1703
R-Squared : -0.0666
MAPE      : 18.35%
```

Figure 5.2.10: Random Forest Performance Analysis

Table 5.2.1: Performance Metrics Comparison between Linear Regression and Random Forest

Metrics	Linear Regression	Random Forest
MAE	0.1154	0.1204
RMSE	0.1649	0.1703
R-Squared	0.0000	-0.0666
MAPE	17.90%	18.35%

Based on the performance output results obtained from running the code on Kaggle as extracted from Figure 5.2.9 and Figure 5.2.10 , the Linear Regression model recorded an MAE value of 0.1154 and an RMSE of 0.1649, which is slightly lower than the MAE (0.1204) and RMSE (0.1703) values by the Random Forest model. Although the difference in error between the two is very small, the R^2 value for Linear Regression of 0.0000 indicates the model's indicates an inability to capture meaningful variance in the target variable in the data, while the negative R^2 value (-0.0666) for Random Forest indicates a worse performance than the average value prediction. This triggers an early indication that both models face great difficulty in providing accurate predictions due to the algorithm's mismatch between the dataset characteristics and the modelling assumptions.

From an error percentage perspective, the MAPE values for both models are in the range of 17.90% to 18.35%, which is considered quite high for critical applications such as electric vehicle battery monitoring which usually requires an error below 10%. Overall, both models do not achieve a satisfactory level of accuracy for use on the selected dataset because the correlation between sensors (Voltage, Current, Temperature) and State of Health (SoH) is close to zero, as evidenced in Figures 5.2.4 and 5.2.8. This weakness of linear and non-linear relationships makes it difficult for the model to identify battery degradation patterns, thus indicating the need for additional processes in model development such as feature engineering in the future.

5.3 RESULTS AND FINDINGS

Based on the performance analysis conducted, it was found that Linear Regression gave slightly better results than Random Forest in terms of average error which is MAE and RMSE which can be seen in Table 5.2.1. Although the difference is very small, Linear Regression recorded an MAE of 0.1154 compared to 0.1204 for Random Forest. This is because the dataset used does not have a strong correlation between the input features (voltage, current, temperature) and the SoH target. In situations where the relationship between variables is very weak or almost random, simpler models such as Linear Regression tend to provide more stable predictions because it only looks for a general trend, a straight line.

On the other hand, the more complex Random Forest model showed worse performance with a negative R^2 value (-0.0666). This is due to the nature of the Random Forest algorithm which tries to build patterns based on data breakdowns (decision trees). When the data correlation is very low, Random Forest tends to experience overfitting problems or be confused by noise in the dataset, causing its predictions to be less accurate than the basic linear model. Overall, these findings prove that the complexity of an algorithm does not guarantee higher accuracy if the raw data does not have significant relationships with each other.

In theory, Random Forest should provide better performance than Linear Regression due to its ability to capture complex nonlinear relationships, the strength of the ensemble technique that combines 100 decision trees through bootstrapping, and its higher robustness against outliers. However, in this project, its performance deteriorated with a negative R^2 value (-0.0666) due to the lack of strong correlation between the input features and SoH. The inability of this model is also clearly visible in Figure 5.2.5 where the prediction points are scattered without direction, proving that without quality input features, complex algorithms such as Random Forest become too sensitive to small variations compared to the stability of the basic linear model.

In conclusion, although Random Forest has the advantage of nonlinear flexibility, the quality of the dataset with almost zero correlation and a large gap between 0.6-0.9 has caused this complex algorithm to be confused by noise, making it less efficient compared to the stability of the basic linear model.

5.4 IDENTIFICATION OF LIMITATIONS OF THE SOLUTION

There are several key constraints that have been identified throughout the implementation of this project that contribute to the low level of model accuracy:

1. **Lack of Feature Correlation (Technical Constraint):** The biggest technical constraint is the quality of the dataset itself. As shown in the Heatmap, the correlation between voltage, current, and temperature sensors with battery health (SoH) is close to zero. This indicates that the physical parameters used are insufficient to describe the battery chemistry degradation in depth. Without additional features, the model will struggle to make accurate predictions.
2. **Hardware and Sensor Constraints:** In real-world situations, the accuracy of the data depends entirely on the quality of the sensors used. Temperature or voltage sensors with high tolerances or electromagnetic interference in the vehicle hardware can produce noisy data.
3. **Time and Data Diversity Constraints:** The dataset used may not cover a wide range of driving scenarios or extreme environmental conditions. Battery health is influenced by long-term factors, and the absence of historical data spanning many years limits the model's ability to effectively learn true degradation trends.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 SUMMARY

Overall, this project tested two AI models, which are Linear Regression and Random Forest, to predict the health of electric vehicle batteries. The results of the study showed that both models have limitations in providing accurate predictions due to the still high mean error (MAPE) values. In fact, Random Forest is a very powerful model and is much better than Linear Regression because of its ability to understand complex and irregular data patterns. However, in this project, its performance was severely affected due to the large gap in data quality, where sensor information (voltage, current, temperature) did not have a strong relationship with battery health. Therefore, clean data quality and accurate information selection are much more important than using complex algorithms alone. The final verdict of this project shows that the two tested AI models are not yet effective enough to be used in real-world applications as long as more relevant input features are not added to the dataset.

6.2 FUTURE WORK

To improve the accuracy of future predictions, several improvement proposals have been identified to overcome the weaknesses of the existing models. The main step that needs to be taken is to overcome the data gap by adding deeper technical features. This is very important because a good model such as Random Forest requires input parameters that have a stronger correlation with battery health to function excellently. In addition, the use of Deep Learning models such as Long Short-Term Memory (LSTM) is highly recommended since the time-series nature of battery data, this algorithm is able to remember the history of past usage and charging to produce much more accurate predictions than regression models. Finally, the provision of a larger dataset that covers a variety of driving situations and extreme environments is very critical. With more diverse data, the AI model will have a wider experience to learn from, thus improving the resilience and accuracy of predictions in real-world applications.

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